

## CORE ENGINEERING PROJECTS

### Sysmora

LEAD BACKEND ENGINEER & CREATOR

**Overview:** A B2B SaaS platform designed for multi-channel inventory synchronization and real-time financial data consolidation. The system aggregates fragmented transaction data from multiple external e-commerce marketplaces and offline POS terminals into a single unified ledger.

**Objective:** Prevent multi-channel stock allocation conflicts across marketplaces and automate financial reconciliation using a robust background data synchronization architecture.

**Tech Stack:** PHP    Laravel Framework    PostgreSQL    Redis (Queue & Cache)    REST APIs    Docker

**Attachments:** Database ERD for multi-tenant isolation, Redis queue worker configuration, webhook handler documentation.

**SPECIFIC CONTRIBUTIONS & MEASURABLE RESULTS**

- **Async Data Sync Engine:** Engineered a background synchronization layer using Redis-backed Laravel Queues, isolating incoming marketplace webhook traffic spikes to protect the application from HTTP request timeouts.
- **Idempotency Layer:** Implemented a custom data reconciliation pipeline to ensure all ledger updates are processed idempotently, preventing duplicate entries caused by automatic network retries.
- **Performance Metrics:** Achieved sub-100ms processing time for local data updates and eliminated stock double-booking gaps in concurrent automated testing scenarios.

### Intelligent Text-to-SQL Analytics Engine

R&D BACKEND & AI ENGINEER

**Overview:** An experimental middleware engine designed to democratize internal data access. It translates natural language business questions into optimized SQL queries ready for execution against relational databases.

**Objective:** Bridge the gap between non-technical stakeholders and complex relational data schemas, eliminating the need for manual ad-hoc query writing by operations teams.

**Tech Stack:** Golang    LLM APIs    Prompt Engineering    PostgreSQL    Context Injection

**Attachments:** Dynamic schema-mapping architecture blueprint, system prompt configuration, SQL validation suite test logs.

**SPECIFIC CONTRIBUTIONS & MEASURABLE RESULTS**

- **Context-Aware Schema Injection:** Built an automated context assembly pipeline that sanitizes and maps database schema metadata for safe injection into the LLM context window without leaking sensitive user row data.
- **Strict Prompt Engineering:** Formulated standardized system instructions and few-shot examples that constrain the LLM to produce pure SQL code output, preventing model hallucinations or markdown formatting wrappers.
- **Execution Validation Layer:** Built a dry-run validation middleware that checks generated queries against destructive commands (DROP, DELETE, UPDATE) before executing read-only operations on the target database.

## Official Tourism Website - Mojokerto City

### BACKEND DEVELOPER

**Overview:** An official regional tourism portal built to digitize information on cultural landmarks, local culinary spots, event calendars, and travel itineraries for the Mojokerto City Government.

**Objective:** Deliver a stable, high-scale platform capable of handling public traffic surges during regional cultural events, while providing a reliable administration panel for content managers.

**Tech Stack:** PHP Laravel Framework MySQL RESTful APIs AdminPanel Integration

**Attachments:** Public deployment logs, REST API endpoint collection, content management user manual.

#### SPECIFIC CONTRIBUTIONS & MEASURABLE RESULTS

- **Content & Media APIs:** Developed high-performance RESTful endpoints to serve structured localization data, tourist attraction geographic coordinates, and multimedia content to the front-end interface.
- **Secure Management Portal:** Built an intuitive administration backend enabling non-technical government operators to safely update historical texts, upload high-resolution image assets, and publish upcoming event bulletins.
- **Query & Asset Optimization:** Applied database caching for static regional tourism assets, reducing repeated hits to the primary database and minimizing per-request memory allocation.

## Enterprise Web & Internal Operations Platforms

### FULL-STACK WEB DEVELOPER (PT. ANDROMEDIA INDONESIA)

**Overview:** Development and optimization of data-intensive corporate web applications and internal workflow automation tools serving large-scale client needs as a replacement for legacy manual procedures.

**Objective:** Design secure and robust backend REST APIs capable of handling complex business workflows, large relational datasets, and multi-tiered access control.

**Tech Stack:** Typescript PostgreSQL PrismaORM RESTful APIs

**Attachments:** API documentation contracts (Swagger/OpenAPI), database optimization logs, corporate compliance reports.

#### SPECIFIC CONTRIBUTIONS & MEASURABLE RESULTS

- **Eloquent Performance Tuning:** Optimized large-scale data aggregation endpoints using eager loading and chunking techniques, preventing memory overflow when processing large analytics reports.
- **Database Schema Refactoring:** Overhauled a client's legacy database schema, applied strategic indexing, and removed redundant subqueries, resulting in reduced CPU utilization during peak hours.
- **Performance Metrics:** Cut query execution time by up to 40% on the primary transactions table, significantly reducing daily support complaint tickets from users related to system timeout issues.

## Distributed Infrastructure R&D Homelab

### SYSTEMS & DEVOPS ENGINEER

**Overview:** An internal bare-metal virtualization environment used to rigorously test high-availability configurations, infrastructure orchestration, system network isolation, and microservices lifecycle management before deploying to public cloud.

**Objective:** Serve as a zero-cloud-cost local testing ground for benchmarking the resilience of stateful services (such as PostgreSQL clusters and Redis caching) against compute failure and network partition scenarios.

**Tech Stack:** Proxmox VE Ubuntu Server Docker / Docker Compose LXC Containers Linux Networking

**Attachments:** Docker Compose automation repository, internal network topology map, hardware utilization monitoring metrics.

#### SPECIFIC CONTRIBUTIONS & MEASURABLE RESULTS

- **Hypervisor Virtualization:** Managed a Proxmox VE node cluster, dynamically balancing compute resource allocation between lightweight LXC containers and full Virtual Machine infrastructure.
- **Rapid Stack Automation:** Composed modular Docker Compose scripts for instant workspace initialization, cutting provisioning time for isolated ready-to-use relational databases down to seconds.
- **Performance Baseline:** Maintained a 99.9% uptime track record for the testing environment, establishing a reliable hardware performance baseline for use as a backend application benchmark reference.